SHISTATA SUBEDI

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EDUCATION

University of Oregon, Eugene, OR — Ph.D. in Computer Science — Sep 2024 – Present
Advisor: Prof. Suyash Gupta

Vellore Institute of Technology, Tamil Nadu, India — B.Tech. in Computer Science and Engineering — May 2019 – Apr 2023

PUBLICATIONS

Smaran: Serving Authenticated Time-Travel Queries. Asim Nepal*, **Shistata Subedi***, Shubham Mishra*, Suyash Gupta, Aniket Kate, Hein Meling. *SOSP 2026.* [Paper] (*equal contribution)

Samurai: Slash Your Decentralized Storage. **Shistata Subedi**. *Middleware 2025, Doctoral Symposium.* [Paper]

RESEARCH EXPERIENCE

Graduate Researcher, University of Oregon, Eugene, OR — Sep 2024 – Present

Smaran — *Authenticated time-travel queries (SOSP 2026)*

- Co-first author of an authenticated data structure that answers historical queries with a constant number of fixed-size proofs, independent of query range or history length.
- Built Go prototypes, extended Geth with Ethereum-state replay and indexing, and evaluated on key transparency and 2.6M Ethereum blocks: 5.7× throughput over CONIKS/OPTIKS and up to 589× over Merkle and Verkle trees.

HotSwap — *Workload-adaptive authenticated storage (3rd place, ACM SRC at SOSP 2026)*

- Reducing proof-maintenance cost in authenticated storage under skewed update workloads; ~3× lower modeled witness-maintenance work on historical workloads.

Cry-for-Help — *Recovering from metastable failures*

- Studying how distributed services can detect and escape metastable overload; built multi-node failure experiments with workload generation and resource throttling.

INDUSTRY & TEACHING EXPERIENCE

Software Engineer, Leapfrog Technology, Kathmandu, Nepal — Apr 2023 – Jun 2024

- Designed and optimized backend services and REST APIs in Java and Micronaut, improving response times by 30%.
- Built features for Aspen Biosciences' drug-discovery platform: experiment tracking, chemistry workflows, and lab inventory.
- Streamlined delivery workflows, cutting project timelines by 20%; led annotation workflows that raised labeling accuracy by 25%.

Teaching Assistant, University of Oregon, Eugene, OR — Sep 2024 – 2025

- Led web programming lectures and hands-on debugging labs for 140+ students.

PROJECTS

MiniSpanner — Sharded, replicated database on Apache ResilientDB, with PBFT within shards and 3PC/Proof-of-Execution across shard leaders. [Code]

HONORS & AWARDS

3rd Place (Graduate), ACM Student Research Competition, SOSP — HotSwap [Winners] — 2026
Lokey Award for Graduate Excellence, University of Oregon — 2024 – 2025
Frank Vignola Microgrant, Eugene–Kathmandu Sister City Association — 2025
COMPEX Full-Ride Scholarship, Embassy of India — 2019 – 2023

TALKS & SERVICE

Presentations: SRC poster, SOSP 2026 · Doctoral Symposium, Middleware 2025 · Poster, USENIX OSDI 2025
Service: Reviewer, IEEE TDSC · Shadow Reviewer, ACM IMC 2025 · Artifact Evaluation Committee, Middleware 2025

SKILLS

Languages: C++, Go, Rust, Python, Java, JavaScript, TypeScript
Systems & tools: Linux, Git, Docker, CloudLab, Ethereum/Geth, PebbleDB, MySQL, MongoDB, Micronaut
Areas: Distributed consensus, replication, storage systems, authenticated data structures, cryptographic commitments